

Random transverse and longitudinal field Ising chains

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I. INTRODUCTION

II. THE MODEL AND KNOWN SDRG RESULTS

We consider the random bond Ising chain in the presence of random transverse and longitudinal fields, defined by the Hamiltonian:

$$\hat{H} = - \sum_{i=1}^L J_i \sigma_i^z \sigma_{i+1}^z - \sum_{i=1}^L \Gamma_i \sigma_i^x - \sum_{i=1}^L h_i \sigma_i^z, \quad (1)$$

Here the $\sigma_i^{x,z}$ are Pauli matrices at site i and we use periodic boundary conditions: $\sigma_{L+1}^z \equiv \sigma_1^z$. The nearest neighbour couplings are ferromagnetic, $J_i > 0$ and random, the transverse fields, $\Gamma_i > 0$ are random, too. The distribution of the longitudinal fields is symmetric: $p(h) = p(-h)$. Throughout the paper we used the following box-like distributions in the calculations:

$$\begin{aligned} \pi(J) &= \begin{cases} 1 & \text{for } 0 < J \leq 1, \\ 0 & \text{otherwise.} \end{cases} \\ \pi_2(\Gamma) &= \begin{cases} 1/\Gamma_0 & \text{for } 0 < \Gamma \leq \Gamma_0, \\ 0 & \text{otherwise.} \end{cases} \\ p(h) &= \begin{cases} 1/h_0 & \text{for } -h_0/2 \leq h \leq h_0/2, \\ 0 & \text{otherwise.} \end{cases} \end{aligned} \quad (2)$$

III. CORRELATION FUNCTIONS

A. The classical limit: $\Gamma_0 = 0$

The longrange order in the system is characterized by the average correlation function

$$C(r) = \overline{\langle \sigma_i \sigma_{i+r} \rangle} \quad (3)$$

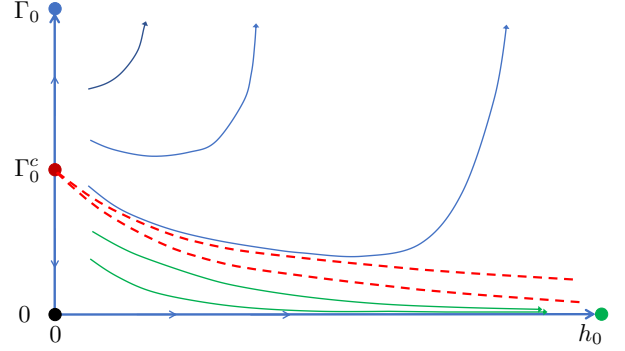


FIG. 1. Schematic RG phase-diagram of the model in Eq.(1) in the thermodynamic limit using the parameters of the box-like distribution in Eq.(2). At $h_0 = 0$ there are two trivial fixed-points: one at $\Gamma_0 = 0$ and indicated by a black circle which controls the quantum ordered phase and another at $\Gamma_0 \rightarrow \infty$ and indicated by a blue circle which controls the quantum disordered phase. These are separated by a non-trivial IDFP denoted by a red circle and located at Γ_0^c . In the classical limit $\Gamma_0 = 0$, we have the random-field Ising model, which has a classical disordered phase for any value of $h_0 > 0$ and its properties are controlled by a fixed-point at $h_0 \rightarrow \infty$ and indicated by a green circle. For general values of the parameters the system has no long-range order and the RG trajectories are attracted by a set of disordered fixed points, in which the couplings are negligible and the disorder is either due to random quantum fluctuations (illustrated by blue trajectories), or due to classical random field effects (illustrated by green trajectories). Between the two regimes there is a cross-over region, the borders of which are illustrated by dashed red lines and which are denoted by $\Gamma_s^+(h_0) > \Gamma_s^-(h_0)$. In the vicinity of the IDFP the cross-over region is sharp, $\lim_{h_0 \rightarrow 0} [\Gamma_s^+(h_0) - \Gamma_s^-(h_0)] = 0$ and the local separatrix defines the relevant scaling direction.

and we choose the two points of reference to be separated by the maximal distance, $r = L/2 - 1$. Here we use the notation:

$$\tilde{C}_L(h) \equiv C(L/2 - 1) \quad (4)$$

We also calculated estimates for the correlation length, ξ , through two-point fit:

$$\xi = \frac{L_2 - L_1}{\ln \tilde{C}_{L_1} - \ln \tilde{C}_{L_2}} \quad (5)$$

In particular we are interested in the limiting behaviour as $p_0 \rightarrow 0$.

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IV. DISCUSSION

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Appendix A: SDRG equations